

Deliverable 3 Report

2) Team Name : Vision Assist Innovators

3) Project Name :

Smart Assistive System

4) Student Names & IDs

- Sarang Prajapati — N01662036
- Krish Patel — N01666556
- Kush Patel — N01657387
- Daksh Rana — N01664095

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6) Project Brief

Smart Assistive System helps visually-impaired users with magnification and emergency assistance. The app integrates camera-based magnifier, SOS one-tap calling, Firebase Auth (email + Google), cloud Feedback capture, and settings with runtime permissions. UI follows a drawer + bottom navigation layout and is production-stable on API 30–36.

7) Signatures & Effort Table

Name	Student Id	GitHub Id	Signature	Effort %
Sarang Prajapati	N01662036	sarangPrajapati2036	S.P	100
Krish Patel	N01666556	KrishPatel6556	KP	100
Kush Patel	N01657387	PatelKush7387	KP	100
Daksh Rana	N01664095	dakshrana4095	DR	100

8) Members Info & Participation (Sprint 3 only)

- **Sarang:** Navigation refactor (drawer + bottom nav), Remember-Me, Registration UI/validation, Google Sign-In integration, Feedback screen wiring to Firebase, About page refresh, project screenshots & PDF assembly.
- **Krish:** Landscape fixes, deprecations cleanup (Activity Result APIs), UI polish, strings cleanup, French updates, SOS API fix, bug-fix PRs.
- **Kush:** MagnifierFragment UI/feature, SOS screen with direct call buttons, Firebase rules tweak, values-fr/strings, Home UI updates, Google Sign-In bugfix.
- **Daksh:** Email/password auth, password auth hardening, Settings with user preferences, Feedback submission logic, sensor data read/display scaffolding, splash Firebase connectivity check.

9) Requirement-by-Requirement Notes

#	Requirement (short)	Status			
10–11	GitHub repo link is valid	✓			
12	Login test with aaa@bbb.com / Admin101!	✓			
13	Other credentials	✓			
14	Show account exists in DB	✓			
15	Auth screen showing multiple users	✓			
16	Sprint 3	✓			
17	Work completed per member	✓			
18–20	≥10 commits/member since Oct 19	✓			
21	Sprint goals	✓			
22–25	Scrum dashboard (Sprint 3 only)	✓			
26–27	≥4 stories; ≥5 tasks each	✓			
28–33	Gantt chart (10+ components w/ assignees)	✓			
33–35	Daily standups (≥3), with 3 answers	✓			
36	Two design principles	✓			

#	Requirement (short)	Status
37	One design pattern	✓
38	Runtime permission	✓
39	Two main features implemented	✓
40	Feedback stored in DB w/ device model	✓
41-42	/docs/deliverable3/ + PDF pushed	✓

9.1 Sprint 3 Goals

- Integrate Google Sign-In with FirebaseAuth
- Implement Customer Feedback (write to Firebase with device model)
- Add runtime permission (Location) with modern API
- Stabilize Magnifier + SOS (core features)
- Finish Gantt and Scrum dashboard artifacts

10) GitHub Repo Link

<https://github.com/KushPatel7387/SmartAssistiveSystem>

11) Login Credentials

- **Email:** aaa@bbb.com
- **Password:** Admin101!
- **Also supported:** Google Sign-In (button on Login screen)

12) Sprint 3 — Work by Member (detail)

Sarang

- Drawer+BottomNav refactor; toolbar About/Help; Logout to drawer
- Remember-Me (SharedPreferences) & Registration validation
- Google Sign-In (client id + SHA-1) and FirebaseAuth bridge
- FeedbackFragment UI polish and Firebase write
- About screen restyle; help text & assets; PDF assembly

Krish

- Fixed deprecated APIs (ActivityResultContracts) & SOS permission flow
- Landscape polish; colors/strings cleanup; small UI fixes
- French strings; removed XML hardcoding
- Bug fixing on Splash/Navigation delays

Kush

- MagnifierFragment UI + feature work; Home UI
- SOSFragment with direct call buttons
- Firebase rules tweak; values-fr/strings
- Google Sign-In bug fix; XML hardcode removal

Daksh

- Email/password auth; password auth hardening
- SettingsFragment with user preferences + permission entry point
- Feedback submission logic and move hardcoded to strings
- Sensor data read scaffolding; splash Firebase connectivity check

13) Commit Counts (since Oct 19)

Member	GitHub Id	Count (>=10)	Evidence)
Sarang Prajapati	sarangPrajapati2036	11	
Krish Patel	KrishPatel6556	10	
Kush Patel	PatelKush7387	10	
Daksh Rana	dakshrana4095	10	

FeedbackFragment hard code move to strings	origin & master	dakshrana4095	10 minutes ago
FeedbackFragment and submission logic		dakshrana4095	14 minutes ago
removed hardcoded values		dakshrana4095	33 minutes ago
SettingsFragment with user preferences		dakshrana4095	36 minutes ago
sensor detection		dakshrana4095	46 minutes ago
data display feature		dakshrana4095	48 minutes ago
password authentication		dakshrana4095	50 minutes ago
added email authentication		dakshrana4095	53 minutes ago
Merge remote-tracking branch 'origin/master'		dakshrana4095	1 hour ago
splash screen with Firebase connectivity check		dakshrana4095	Today 9:58 PM
Made some final touches		sarangPrajapati2036	Today 9:45 PM
Delay fix in Splash Screen		KrishPatel6556	Today 9:02 PM
French texts updated		KrishPatel6556	Today 8:45 PM
Removed hardcoding from XML		sarangPrajapati2036	Today 8:43 PM
UI issue fix		KrishPatel6556	Today 8:39 PM
in SOS fragment removed the deprecated api		KrishPatel6556	Today 8:34 PM
replaced deprecated onRequestPermissionsResult with ActivityResultLauncher		PatelKush7387	Today 8:19 PM
Updated UI of HomeFragment.java		PatelKush7387	Today 8:13 PM
Names and N number added on the top of each java files		KrishPatel6556	Today 8:00 PM
Implemented UI and added feature for SensorFragment.java		PatelKush7387	Today 7:19 PM

Remove hardcoded from Java Class files		PatelKush7387	Today 2:22 PM
fix(firebase-rules) : corrected Feedback node name to uppercase and enable		PatelKush7387	Today 2:13 PM
added French translations strings in values-fr/strings.xml		PatelKush7387	Today 1:51 PM
Updated UI of MagnifierFragment.java and implemented feature		PatelKush7387	Today 1:38 PM
implemented working SOSFragment with direct call buttons for ambulance		PatelKush7387	Today 1:11 PM
Created UI of SOSFragment.java		PatelKush7387	Today 12:57 PM
"fix: implement working Google Sign-In with FirebaseAuth		PatelKush7387	Today 12:21 PM
Registration page UI update		KrishPatel6556	10/31/2025 10:16 PM
Register User is in progress		sarangPrajapati2036	10/31/2025 2:19 AM
Added Feedback fragment		sarangPrajapati2036	10/31/2025 12:49 AM
Added Remember me checkbox, works with dummy account		sarangPrajapati2036	10/31/2025 12:36 AM
changes for Landscape layout		sarangPrajapati2036	10/30/2025 11:53 PM
Merge remote-tracking branch 'origin/master'		sarangPrajapati2036	10/30/2025 11:47 PM
Created Bottom navigation bar, improved the toolbar		sarangPrajapati2036	10/30/2025 11:35 PM
Created Bottom navigation bar, improved the toolbar		sarangPrajapati2036	10/30/2025 11:35 PM
Moved Logout to Nav, About , help in overflow		sarangPrajapati2036	10/30/2025 11:22 PM
Removed X & F from login sc. Added names in .git		sarangPrajapati2036	10/30/2025 10:36 PM

14) Scrum Dashboard — Sprint 3

▼ Story 3: Mobile Application

☐	Task		Owner	Status ⓘ	Due date ⓘ	Priority	
☐	Design login & splash screens	✦ Ⓢ	KP KP	Done	ⓘ Oct-4	Medium	
☐	Implement navigation drawer	✦ 🔔	KP	Done	ⓘ Oct-4	High	
☐	Add settings page (runtime permis...	✦ Ⓢ	KP KP	Working on it	ⓘ Oct 4	Medium	
☐	Add Google sign in	✦ Ⓢ	👤	Done	✓ Nov-2	High	
☐	Test landscape & portrait layouts	✦ 🔔	KP KP	Done	ⓘ Oct-3	Low	
☐	Style fragments using colors.xml an...	✦ Ⓢ	👤 DR	Done	✓ Oct-5	Low	
☐	Design bilingual support (English/Fr...	✦ Ⓢ	👤 DR	Done	✓ Oct-5	Low	
☐	Magnifier features (Camera)	⌂	Preparing print preview		✓ Nov-2	High	Help
☐	SOS , Call feature	✦ Ⓢ	KP	Done	✓	High	

✓ Story 1: Database Setup

☐	Task		Owner	Status ⓘ	Due date ⓘ	Pri
☐	Design DB schema on paper	✦ ⓧ		Done	ⓘ Sep 30	L
☐	Create DB instance on cloud (Fireb...	✦ 🔔	KP	Done	✓ Oct 5	L
☐	Connect Android app to DB	✦ ⓧ	KP	Done	✓ Oct 6	L
☐	Test data insertion/retrieval	✦ ⓧ	KP +2	Working on it	○ Nov 20	H
☐	Test CRUD operations with sample ...	✦ ⓧ	DR	Done	✓ Nov 2	Me
☐	+ Add task					
					Sep 30 - No...	

✓ Story 2: Sensor Integration

☐	Task		Owner	Status ⓘ	Due date ⓘ	Pri
☐	Interface AS7262 Visible Sensor	✦ ⓧ		Working on it	○ Dec 5	L
☐	Interface TSL2591 Light Sensor	✦ 🔔	KP	Not Started	○ Dec 5	L
☐	Test sensor data with Android app ...	✦ ⓧ	KP KP	Working on it	○ Dec 31	L
☐	Calibrate readings, store results in c...	✦ ⓧ	KP	Not Started	○ Dec 1	L
☐	Document calibration constants & ...	✦ ⓧ	Filter board by person d			○ Nov 30
☐	Placing order for sensors	✦ ⓧ	KP +3	Done	ⓘ Oct 4	H
☐	Integrate TCS34725 Color Sensor	✦ 🔔	DR	Not Started	○ Dec 5	L
☐	Integrate VL53L1X Distance Sensor	✦ 🔔	KP	Not Started	○ Nov 15	L

✓ Story 4: Testing & Deployment

☐	Task		Owner	Status ⓘ	Due date ⓘ	Pri
☐	Perform unit testing on main comp...	✦ ⓧ	KP	Not Started	○ Dec 1	Medi
☐	Debug DB & sensor issues	✦ ⓧ	DR	Not Started	○ Dec 5	Low
☐	Package APK for demo	✦ ⓧ	DR	Not Started	○ Dec 5	Medi
☐	Deploy on test device	✦ ⓧ	KP DR	Done	✓ Nov 24	Low
☐	Capture screenshots & prepare de...	✦ ⓧ	DR	Done	✓ Dec 25	Low

✓ Story 5: Business Model Canvas & Gantt

☐	Task		Owner	Status ⓘ	Due date ⓘ	Pri
☐	Create Business Model Canvas	✦ ⓧ	KP KP	Done	ⓘ Oct 4	Low
☐	Build Gantt chart on GrantPro	✦ ⓧ	DR	Done	✓ Oct 5	Hig
☐	Integrate screenshots in PDF	✦ ⓧ	DR	Done	✓ Oct 5	Low
☐	Verify professor access to repo	✦ ⓧ	KP	Done	✓ Oct 5	Low
☐	Review team effort table and signat...	✦ ⓧ	+3	Done	✓ Oct 5	Low

▼ 6 Auth & Security

☐	Task		Owner	Status ⓘ	Due date ⓘ
☐	Add SHA 1, Confiure Authenticatio...	✦ ⬆		Working on it	Nov 8
☐	Lock Realtime DB rules	✦ ⬆		Done	Nov-1
☐	Capture Firebase Auth users screen...	✦ ⬆		Done	Nov-1
☐	capture Readtime Entires	✦ ⬆		Done	Nov-1
☐	+ Add task				

15) Gantt Chart — 10+ Components with Assignees

Smart Assistive System

☆

No status ▼

Gantt chart

Board

List

Calendar

Workload

People

Dashboard

☑

📅

🔍

🔗

⏏

Expand all

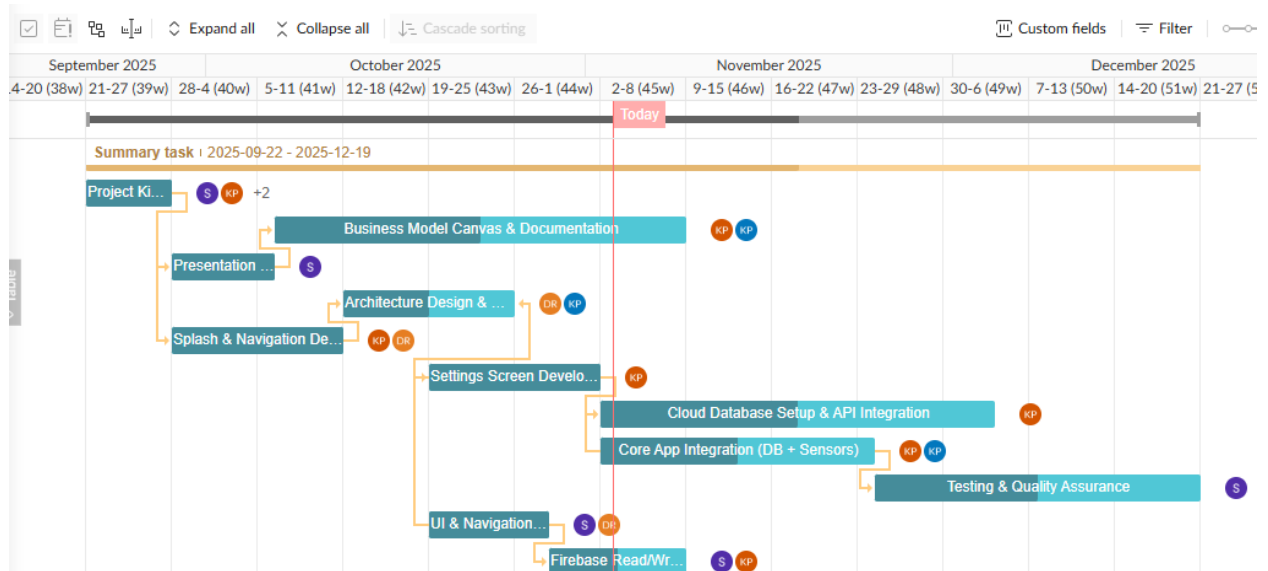
⏏

Collapse all

⏏

Cascade sorting

Task name	Status	Assigned
1 ☐ Summary task		
1.1 Project Kickoff & Requirements	● Done	S KP +2
1.2 Business Model Canvas & Documentation	● In progress	KP KP
1.3 Presentation Preparation & Review	● Done	S Sarang Prajapati
1.4 Architecture Design & GitHub Setup	● In progress	DR KP
1.5 Splash & Navigation Development	● Done	KP DR
1.6 Settings Screen Development	● Done	KP Kush Patel
1.7 Cloud Database Setup & API Integration	● In progress	KP Kush Patel
1.8 Core App Integration (DB + Sensors)	● In progress	KP KP
1.9 Testing & Quality Assurance	● In progress	S Sarang Prajapati
1.10 UI & Navigation Foundation	● Done	S DR
1.11 Firebase Read/Write Demo	● In progress	S KP



16) Daily Standup

Date	Yesterday	Today	Blockers	Attended / Missed
Oct 29	e.g., Finished Login UI polish	Wire Feedback to Firebase	Need SHA-1 added to Firebase	All
Nov 2	Added SOS call flow	Google Sign-In integration	SHA-1 mismatch (fixed)	All
Nov 1	Gantt + Scrum updates	Final screenshots & PDF	Emulator ANR (resolved)	All

17) Two Design Principles (with code)

17.1 Separation of Concerns (SoC)

Where used: `SettingsFragment` keeps UI & permission handling local; DB/auth lives elsewhere.

Snippet

```
permissionLauncher = registerForActivityResult(
    new ActivityResultContracts.RequestPermission(),
    isGranted -> { /* UI feedback only; OS permission logic is separate */
    });
```

Why it's good: UI code doesn't mix with DB/auth; permission request is triggered from a button (no over-permissioning). Easier to test and maintain.

17.2 Single Responsibility Principle (SRP)

Where used: `RegisterActivity.checkAndRegister(...)` only validates + delegates to Firebase Auth/DB; no navigation or unrelated logic mixed in.

```
mAuth.createUserWithEmailAndPassword(email, password)
    .addOnCompleteListener(task -> {
        if (task.isSuccessful()) {
            String uid = mAuth.getCurrentUser().getUid();
            usersRef.child(uid).setValue(new UserModel(name, phone, email,
"regular"));
            finish();
        }
    });
```

Why it's good: Each method has a focused job. Changes in DB schema/auth don't impact the rest of UI.

18) One Design Pattern — Singleton/Facade for Firebase

Added a tiny helper This centralized Auth/DB refs and avoids duplicate initialization.

```
// ca.visionassistinnovators.it.smartassistivesystem.data.FirebaseRefs
public final class FirebaseRefs {
    private static FirebaseAuth auth;
    private static FirebaseDatabase db;
    private static DatabaseReference users;
    private static DatabaseReference feedback;

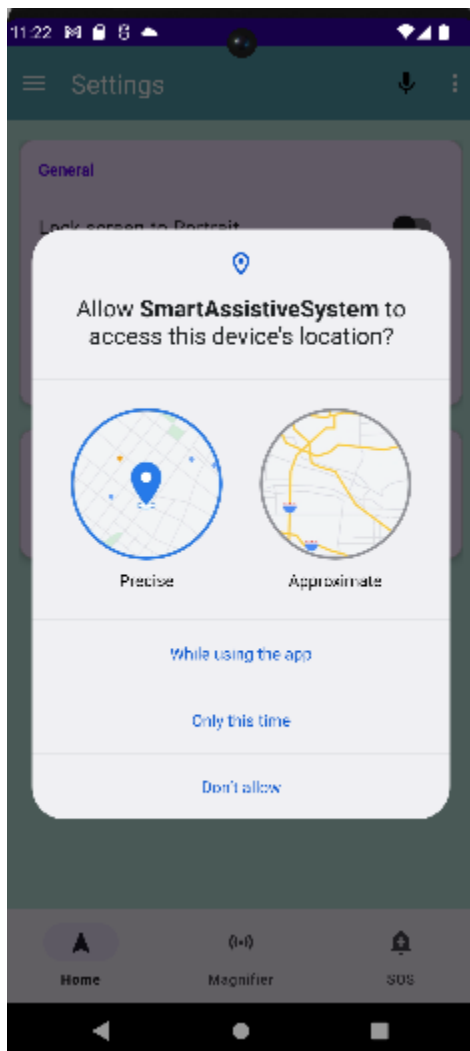
    private FirebaseRefs() {}

    public static synchronized FirebaseAuth auth() {
        if (auth == null) auth = FirebaseAuth.getInstance();
        return auth;
    }
    public static synchronized FirebaseDatabase db() {
        if (db == null) db = FirebaseDatabase.getInstance(
            "https://smartassistivesystem-39072-default-rtdb.firebaseio.com/");
        return db;
    }
    public static DatabaseReference users() {
        if (users == null) users = db().getReference("users");
        return users;
    }
    public static DatabaseReference feedback() {
        if (feedback == null) feedback = db().getReference("feedback");
        return feedback;
    }
}
```

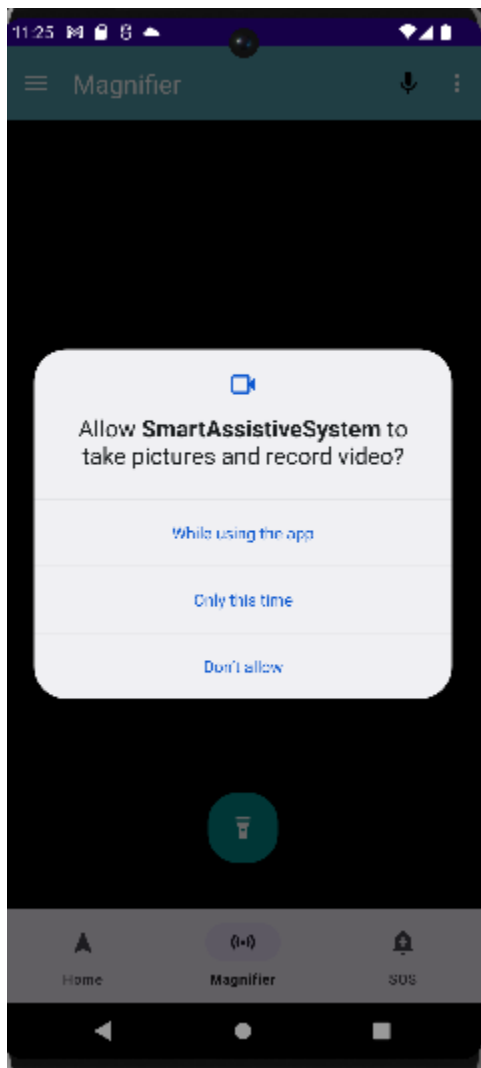
```
}  
}  
.
```

19) Runtime Permission (Location)

- Implemented via **ActivityResultContracts.RequestPermission** in **SettingsFragment**.



Camera access and call



20) Two Main Features

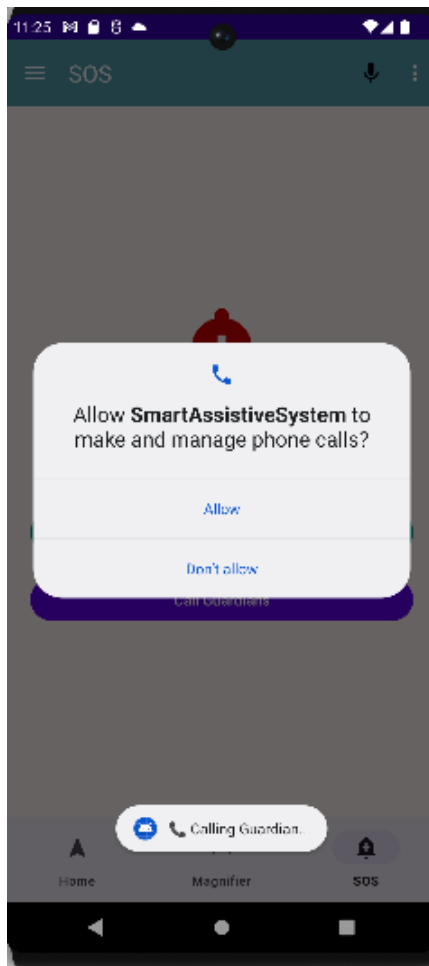
20.1 Magnifier (Camera) — [SS-Magnifier]

Comment: Live camera preview with enlarged content; helps users read nearby text/objects.



20.2 SOS (Calling) — [SS-SOS]

Comment: One-tap call to emergency contacts (runtime `CALL_PHONE` handled via Settings/permission flow).



21) Feedback → Cloud DB

- **UI fields:** name, phone, email, comment, stars
- **Extra captured:** Build.MODEL stored but not shown to user
- **DB path:** /feedback/{autoId}

https://smartassistivesystem-39072-default-rtdb.firebaseio.com/



Feedback



-0d4ybKDKUi01YD0HNDf

comment: "Its very helpful for me."

deviceModel: "Google sdk_gphone64_x86_64"

email: "kp65619@gmail.com"

name: "kush"

phone: "4374219150"

SensorData



LightSensor

latest: 0

timestamp: "2025-11-02 20:09:35"

users



-0csqZQG83QfaiE_x71a

email: "aaa@mad.com"

name: "fsf"

phone: "45554"

Authentication stored Google

Authentication

[Users](#) | [Sign-in method](#) | [Templates](#) | [Usage](#) | [Settings](#) | [Extensions](#)

 The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps. 

 Search by email address, phone number, or user UID

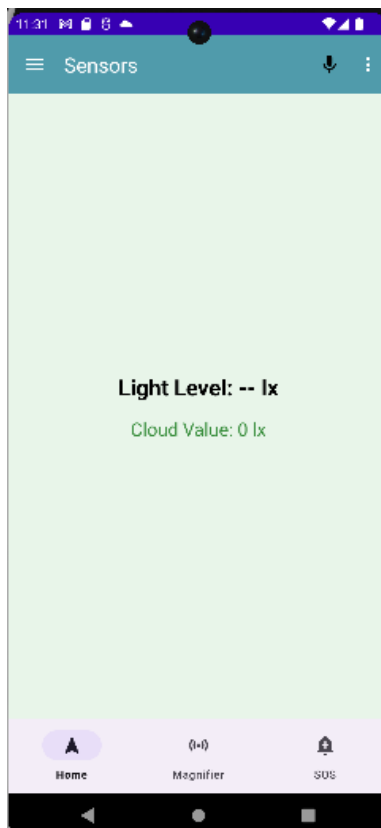
[Add user](#)



Identifier	Providers	Created ↓	Signed In	User UID
kp65619@gmail.com		Nov 2, 2025	Nov 2, 2025	4pPpRT6JGAS3l6HXseJ3w1n...
smartassistivesystem...		Nov 2, 2025	Nov 2, 2025	lTaYibTsbJY4n1xMzFvxJujFg9...
sarang@s.com		Nov 2, 2025	Nov 2, 2025	MdrxwbCKl6bjlCy9QJFWUu9...
worldofcars40@gmail.c...		Nov 2, 2025	Nov 2, 2025	JKTOQDToPida3zWJuD1pwP...

22) Read-From-DB Evidence

Include a small screen (e.g., Alerts/Sensors) that **reads** a list/value from Firebase and displays it.



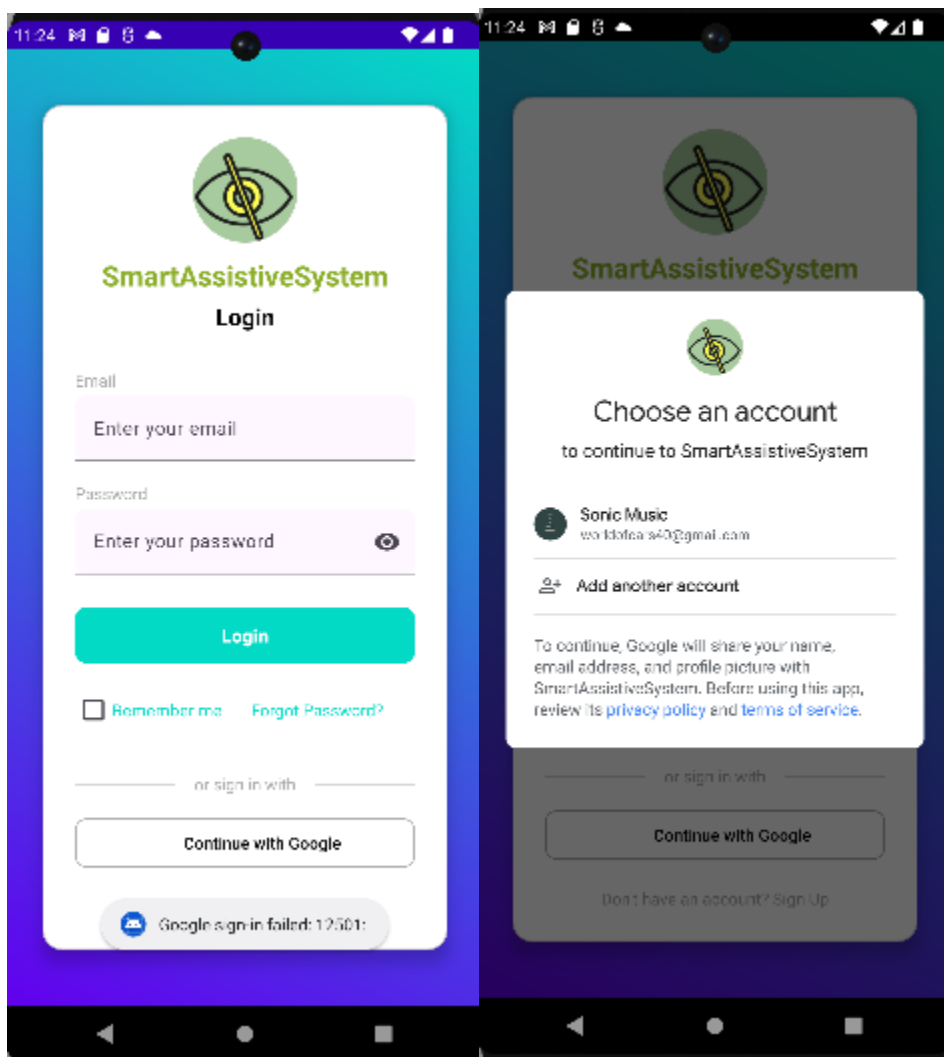
23) SharedPreferences Evidence

- **Remember-Me:** `sas_prefs` with `remember_me=true` and saved email

```
cm9pZDpkYzNiMDdkYWQwM2VjNzdhNmE5NjQ2.xml  </> sas_prefs.xml x  © HomeFragment.java  v
1  <?xml version='1.0' encoding='utf-8' standalone='yes' ?>  Reader Mode
2  <map>
3      <boolean name="remember_me" value="false" />
4  </map>
5
```

Screenshot Index

- [SS-Repo] Repo home
- [SS-Login] Login screen (email + Google)



- [SS-Auth] Firebase Auth users list (email + Google)

Authentication

Users Sign-in method Templates Usage Settings Extensions

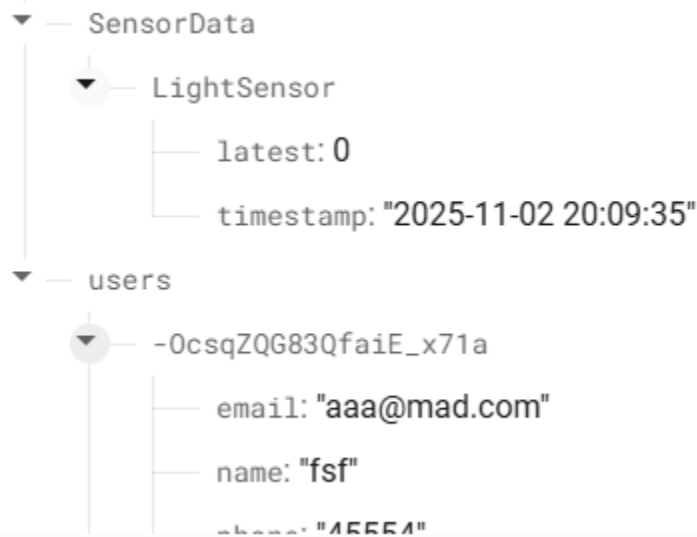
i The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps.

Search by email address, phone number, or user UID

Add user

Identifier	Providers	Created ↓	Signed In	User UID
kp65619@gmail.com		Nov 2, 2025	Nov 2, 2025	4pPpRT6JGAS3l6HXseJ3w1n...
smartassistivesystem...		Nov 2, 2025	Nov 2, 2025	lTaYibTsbJY4n1xMzFvxJujFg9...
sarang@s.com		Nov 2, 2025	Nov 2, 2025	MdrxwbCKl6bjlCy9QJFWUu9...
worldofcars40@gmail.c...		Nov 2, 2025	Nov 2, 2025	JKTOQDToPida3zWJuD1pwP...

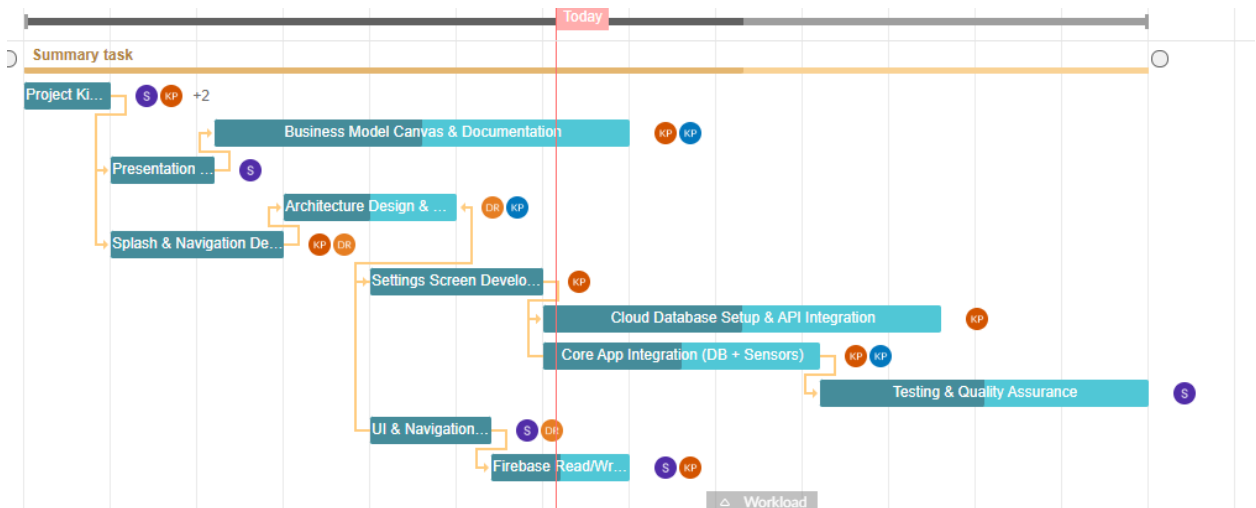
-
- [SS-DB-User] Realtime DB /users/uid with aaa@bbb.com



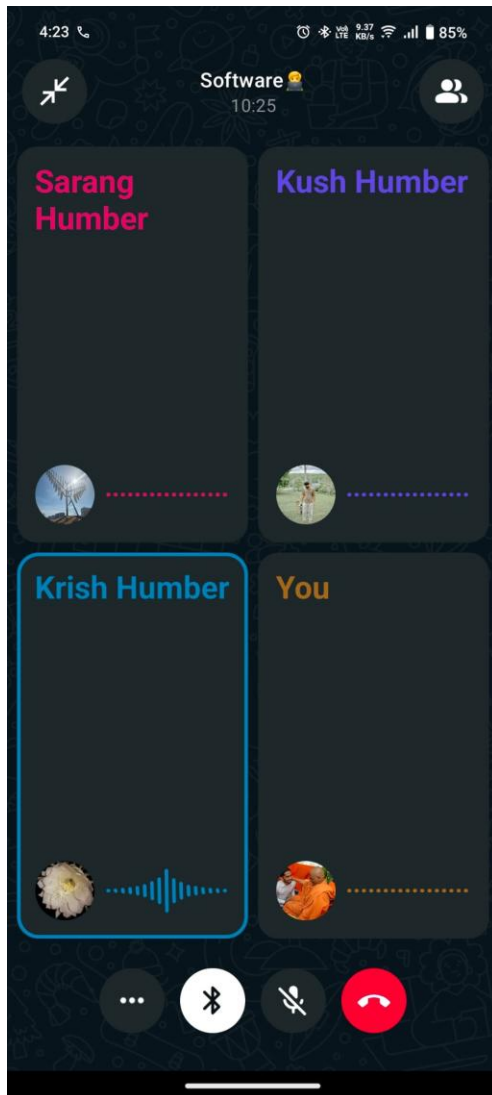
-
- [SS-Scrum-Stories] Sprint 3 board (4+ stories × 5 tasks)

> Story 1: Database Setup 5 Tasks		Status 	Due date Sep 30 - No...	Priority
> Story 2: Sensor Integration expand group S		Status 	Due date Oct 1 - Dec 31	Priority
> Story 3: Mobile Application 10 Tasks		Status 	Due date Oct 3 - Nov 2	Priority
> Story 4: Testing & Deployment 5 Tasks		Status 	Due date Nov 21 - Dec...	Priority
> Story 5: Business Model Canvas & Gantt 5 Tasks		Status 	Due date Oct 1 - 5	Priority

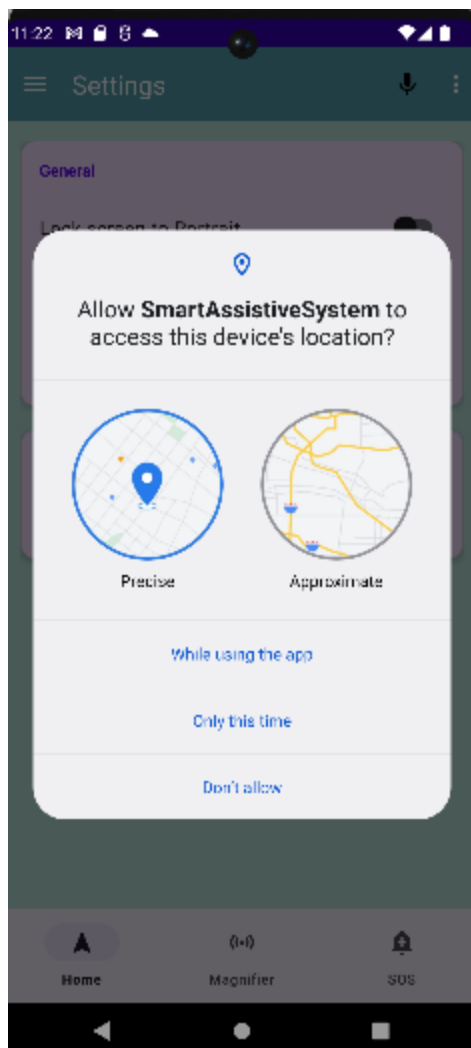
- [SS-Gantt] Final Gantt with assignees

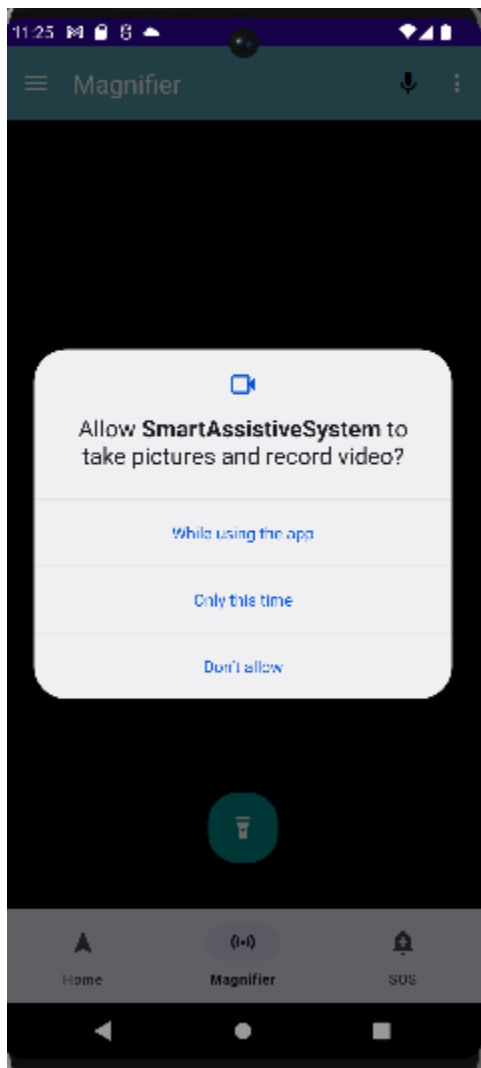


- [SS-Standups] Standup table screenshot



- [SS-PermissionFlow] Permission request flow
- Implemented via **ActivityResultContracts.RequestPermission** in `SettingsFragment`.





- [SS-Magnifier] Magnifier UI



- [SS-SOS] SOS UI

